

Math 107-01
Test 3A: November 4, 2013

Answer all of the following questions. You will have 50 minutes to complete the test. You may not use any outside assistance on this test. You may use a calculator, but you may not program information into the calculator. The use of electronic equipment such as mp3 players, ipods, cell phones and any electronic devices other than your calculator during the test is prohibited. If you have time, check your answers for accuracy. **Show all work. Unsupported answers may not receive credit.** Good luck!

Name: _____

1. A die is rolled and a coin flipped. Find the following (3 points each):
 - (a) The sample space

 - (b) The number of outcomes in the sample space

2. A card is dealt from a standard deck of 52 cards. Find the probability of being dealt each of the following. State your answers as reduced fractions (4 points each):
 - (a) a four

 - (b) a diamond

 - (c) not a diamond

 - (d) the four of diamonds

 - (e) a four or a diamond

3. A die is rolled. Find the following (3 points each):

(a) the probability of rolling a 2

(b) the odds of rolling a 2

(c) the probability of rolling a number greater than 4

(d) the odds of rolling a number greater than 4

4. A jar contains 35 jellybeans. There are twelve blue, eight red, ten yellow, and five green jellybeans. You pick 5 jellybeans without looking. Find the probability of each of the following. Show all work and round your answers to four decimal places (4 points each):

(a) all five are blue

(b) exactly four are blue

(c) at least four are blue

5. (4 points each)

- (a) For a certain factory, the probability of selecting a worker earning a particular hourly wage is as follows:

Hourly wage	\$8.50	\$9.00	\$9.50	\$10.00	\$10.50	\$ 11.00
Probability	15%	20%	25%	20%	15%	5%

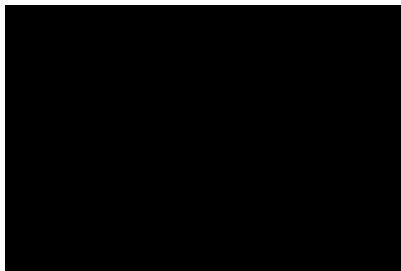
Find the expected hourly wage that a worker at this factory makes. Round to the nearest cent.

- (b) Find the expected value of the following game:
You roll a die. If you roll a 2, 4, or 6, then you win \$15.
If you roll a 3 or 5, then you lose \$10.
If you roll a 1, then you lose \$20.
Round to the nearest cent.

6. In a certain Math 107 class, 75% of the students live on campus. 80% of the students are taking an English course. 60% of the students live on campus and are taking an English course.

Let $L = \{\text{live on campus}\}$ and $E = \{\text{taking an English course}\}$.

Construct a Venn diagram illustrating the situation and find $p(L \cup E)$. (5 points)



$$p(L \cup E) = \underline{\hspace{2cm}}$$

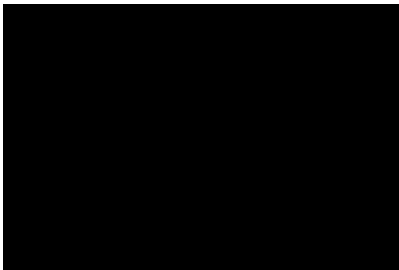
7. A and B are two events for a sample space. $p(A) = 0.58$ and $p(B) = 0.29$. You must show all work to receive full credit. (3 points each):

(a) If $p(A \cap B) = 0.16$, what is $p(A \cup B)$?

(b) If A and B are mutually exclusive events, what is $p(A \cup B)$?

8. The probability that the new library is finished in time for the Summer Festival is 55%. If the library is finished for the Summer Festival, the probability it will have a roof that leaks is 25%. If the library is not finished in time for the Summer Festival, the probability that it will have a roof that leaks is 10%.

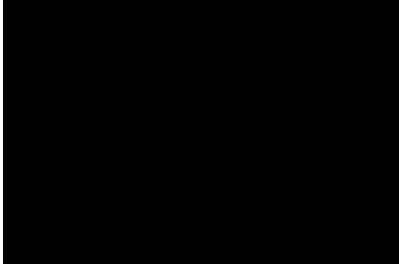
Construct a tree diagram, including the probability of each limb. (10 points)



9. Of the students who are Biology majors, 55% are taking Calculus and 65% are taking Ecology. 30% of the Biology majors are taking both courses.

Let $C = \{\text{taking Calculus}\}$ and $E = \{\text{taking Ecology}\}$.

The Venn diagram is given below.



Jack is a Biology major. Calculate the following probabilities (3 points each):

- (a) Jack is not taking Calculus
- (b) Jack is taking Calculus and Ecology
- (c) Jack is taking Calculus or Ecology
- (d) Jack is not taking Calculus and not taking Ecology
- (e) Jack is not taking Calculus and is taking Ecology
- (f) Jack is taking Calculus given that he is taking Ecology
- (g) Jack is taking Ecology given that he is taking Calculus

On my honor, I have neither given nor received any academic aid or information that would violate the Honor Code of Mars Hill University. _____ (please sign)